Python king is in process

how to use jupiter notebook

Basic of python

First program

In [2]:
```
print(2+3)
print("hello world")
print("i am coming")
```

```
5
hello world
i am coming
```

2_Operators

In [3]:
```
print(2+2)
print(2-3)
print(2/3)
print(2**3)
print(3%2)
print(2**3/3-4+2%5)
```

```
4
-1
0.6666666666666666
8
1
0.6666666666666665
```

PEMDAS

parantesis exponential multiplation division addition subtration

3_Strings

In [4]:
```
print(2+3)
print("hello world")
print("i am coming")
```

```
print('hello world')
print('''Hello world2''')
```

```
5
hello world
i am coming
hello world
Hello world2
```

4_Comments "ctrl+/"

In [5]:
```
# print(2+3)
# print("hello world")
# print("i am coming") press this to comment out ctrl+/
```

5_Variables

"# Rule of variable

1- variable must be number letter or underscore

2- do not start with number

3- spaces are not allowed

4- do not use function name like type print mean median

5- short and descriptive

6- case sensitivity keep an eye on upper case or lower case suggestion is lower case use normally"

In [6]:
```
x=5
y= "i am going to be the python king soon"

print(x)
print(y)
```

```
x=15
print(x)
x=x+10
print(x)
type(x)
print(type(x))
print(type(y))

fruit_basket=8
type(fruit_basket)
print(type(fruit_basket))
print(fruit_basket)
fruit_basket="mangoes" , "oranges"
type(fruit_basket)
print(fruit_basket)
print(type(fruit_basket))
```

```
5
i am going to be the python king soon
15
25
<class 'int'>
<class 'str'>
<class 'int'>
8
('mangoes', 'oranges')
<class 'tuple'>
```

6_Input Variables

In [7]:
```
fruit_basket=input("what is your favourite fruit")
print(fruit_basket)
```

```
mango
```

input function 2nd stage

In [8]:
```
name=input("what is your name")
greetings="Hello!"
print(greetings,  name)
```

```
Hello! asim
```

another way of input 2nd stage

In [9]:
```
name=input("what is your name")
print("hello!",  name)
```

```
hello! asim
```

3rd stage of input variable

In [10]:
```
name=input("What is your name?  ")
age=input("how old are you?"  )
```

```
greetings=("Hello!")
print(greetings, name , "you are still young" )
```

```
Hello! asim you are still young
```

4th stage of input variable

In [11]:
```
name=input("What is your name?  ")
age=input("how old are you?"  )
greetings=("Hello!")
print(greetings, name , age )
```

```
Hello! asim 34
```

07_logical operator are either true or false, yes or no, 0 or 1

equal to ==

not equal to !=

less than <

greater than >

less than and equal to <=

greater than and equal to >=

is 4 equal to 4?m

In [12]:
```
print(4==4)
print(4!=4)
print(4>3)
print(4<3)
print(4<=6)
print(4>=6)

# hammad_age=4
# school_age=5
# print(hammad_age>=school_age)
```

```
True
False
True
False
True
False
```

logical operator applications

In [13]:
```
hammad_age=4
school_age=5
print(hammad_age>=school_age)
```

```
False
```

Input parameter and conditional

In [14]:
```
hammad_age=input("how old are you?")
print(type(hammad_age))
hammad_age=int(hammad_age) #above input is string to convert to int use int functio
print(type(hammad_age))
school_age=5
print(hammad_age>=school_age)
```

```
<class 'str'>
<class 'int'>
True
```

8_Type COversion

In [15]:
```
x=10        #int
y=10.2      #float
z="hello"   # str

print(type(x))
print(type(y))
print(type(z))
```

```
<class 'int'>
<class 'float'>
<class 'str'>
```

implicit type conversion

In [16]:
```
x=x*y
print(x,"type of x is :  ", type(x))
```

```
102.0 type of x is :   <class 'float'>
```

explicit type conversion

In [17]:
```
age=input("what is your age?  ")
print(type(age))
age=int(age)
print(type(age))
print(type(int(age)))
print(age,type(int(age)))

print(type(age))
age=int(age)
print(type(age))
print(type(float(age)))
```

```
name=input("what is your name?  ")
print(name, type(str(name)))
```

```
<class 'str'>
<class 'int'>
<class 'int'>
33 <class 'int'>
<class 'int'>
<class 'int'>
<class 'float'>
asim <class 'str'>
```

9_IF_ELIF_ELSE

can use as many as elif statements

In [19]:

```
age_at_school=5
aqeel_age=4
# question is can aqeel join school?

if age_at_school==aqeel_age:
    print("Aqeel can join school")
else:
    print("aqeel cannot go to school")


age_at_school=5
aqeel_age=6
# question is can aqeel join school?

if age_at_school==aqeel_age:
    print("Aqeel can join school")
else:
    print("aqeel cannot go to school")

age_at_school=5
aqeel_age=2
# question is can aqeel join school?

if age_at_school==aqeel_age:
    print("Aqeel can join school")
elif aqeel_age > age_at_school:
    print("aqeel should join higher classes")
elif aqeel_age==2:
    print("you should take care of aqeel")
else:
    print("aqeel cannot go to school")
```

```
aqeel cannot go to school
aqeel cannot go to school
you should take care of aqeel
```

10_Functions

In [20]:
```
print("Asim is going to be python king soon")
print("Asim is going to be python king soon")
print("Asim is going to be python king soon")
print("Asim is going to be python king soon")
print("Asim is going to be python king soon")
print("Asim is going to be python king soon")
```

```
Asim is going to be python king soon
Asim is going to be python king soon
Asim is going to be python king soon
Asim is going to be python king soon
Asim is going to be python king soon
Asim is going to be python king soon
```

defining own function type 1

In [21]:
```
def print_asim():
    print("Asim is going to be python king soon")
    print("Asim is going to be python king soon")
    print("Asim is going to be python king soon")

print_asim()
```

```
Asim is going to be python king soon
Asim is going to be python king soon
Asim is going to be python king soon
```

defining a function type 2

In [22]:
```
def print_asim():
    text=" Asim is going to be python king soon"
    print(text)
    print(text)
    print(text)

print_asim()
```

```
Asim is going to be python king soon
Asim is going to be python king soon
Asim is going to be python king soon
```

type 3

In [23]:
```
def print_asim(text):
    print(text)
    print(text)
    print(text)
print_asim("Asim is going to be the champ of python")
```

```
Asim is going to be the champ of python
Asim is going to be the champ of python
Asim is going to be the champ of python
```

type 4 using if elif else

In [24]:
```
def school_function(age, text):
    if age==5:
        print("hammad can join the school")
    elif age>5:
        print("hammad can join higher classes")
    else:
        print("hammad is still a baby")

school_function(10, "hammad")
```

```
hammad can join higher classes
```

type 5

In [25]:
```
def school_function(age):
    if age==5:
        print("hammad can join the school")
    elif age>5:
        print("hammad can join higher classes")
    else:
        print("hammad is still a baby")
school_function(10)
```

```
hammad can join higher classes
```

defining a function of future

In [26]:
```
def future_age(age):
    new_age=age+20
    return new_age
    print(new_age)

future_predicted_age=future_age(18)
print(future_predicted_age)
```

```
38
```

11 while and for loops two main types

discussing while loop

In [27]:
```
x=0
while x<5:
    print(x)
```

```
    x=x+1

days= "mon","tue","wed","thu","fri","sat","sun"
for d in days:
    if (d=="fri"):continue # skips d or fri
    print(d)
```

```
0
1
2
3
4
mon
tue
wed
thu
sat
sun
```

For loop

In [28]:
```
for x in range(5,10):
    print(x)
```

```
5
6
7
8
9
```

array in loop function with break

In [29]:
```
days= "mon","tue","wed","thu","fri","sat","sun"
for d in days:
    if (d=="fri"):break #loop breaks on d or friday
    print(d)
```

```
mon
tue
wed
thu
```

continue fuction

In [30]:
```
days= "mon","tue","wed","thu","fri","sat","sun"
for d in days:
    if (d=="fri"):continue # skips d or fri
    print(d)
```

```
mon
tue
wed
thu
sat
sun
```

this is used to recall the fuction made by authors

notes

important libraries

numpy , pandas , etc

In [31]:
```
import math
print("the value of pi is ",math.pi)
```

```
the value of pi is  3.141592653589793
```

another library call

In [32]:
```
import statistics
x=[100,200,300,400,500,600]
print(statistics.mean(x))
```

```
350
```

12troubleshooting

print(i am going to be the king of python)

syntax error are those where you missed the punctuations ie python language error

print(25/0) #zerodivision error called runtime error ie mathematical error

In [33]:
```
name= "Asim"
print("hello name")  # symentic error difficult to trace
```

```
hello name
```

In [ ]:
```

```